CSE2023 Midterm Exam

(1) $A=\{1,2,3\}$ $A\cap B = A\cap C = \{1,2\}$

$B=\{1,2,4\}$ $A\cup B = \{1,2,3,4\}$ a) if $B\neq C$ + $A\cup B \neq A\cup C$

$C=\{1,2,5\}$ $A\cup C = \{1,2,3,5\}$ so $B=C$ must be true

b) $A\cap B = A\cap C = \{1,2\}$ but $B\neq C$ so this statement is false

(2) a) $f(m,n)=2^m\cdot 3^n = f(k,\ell) = 2^k\cdot 3^\ell \rightarrow 2^{m-k} = 3^{\ell-n}$

2 and 3 prime numbers then $2^{m-} = 3^{-n} \leftrightarrow m-k=0) \wedge (\ell-n=$

$(m-k=0 \rightarrow m=k) \wedge (\ell- =0 \rightarrow \ell=n)$ s is one-to-one

$[(m\in N)\wedge(n\in N)] \rightarrow f(m,n)\neq 25$ f is not onto

b) $u(n)=n+1$ $uos = 2n+2$ $sos = 2(2n+1)+1 = 4n+3$

$s(n)=2n+1$

(3.) a) 1 b) 0 c) 1 d) 0 e) 1 f) 0 g) 0 h) 0
i) 1 j) 0

(4.) $p \to (q \wedge r)$, $r \to s$ and $\neg(q \wedge s)$

$p \to (q \wedge r) \equiv (p \to q) \wedge (p \to r) \equiv (\neg p \vee q) \wedge (\neg p \vee r) \equiv \neg p \wedge (q \vee r) \equiv \neg p$

$r \to s \equiv \neg r \vee s$

(5.)

p	q	r	$(\neg q \wedge r)$	$(q \wedge r)$	$(p \wedge r)$	$(\neg p \wedge (\neg q \wedge r))$	$(a \vee b \vee c)$
0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0
0	0	1	1	0	0	1	1
1	1	1	0	1	1	0	1
1	0	1	1	0	1	0	1
0	1	1	0	1	0	0	1
1	1	0	0	0	0	0	0

(a) (b) (c)

$(\neg p \wedge (\neg q \wedge r)) \vee (q \wedge r) \vee (p \wedge r) \equiv r$

(6.) $\bar{a}$ is inverse of 7 mod 19

$\bar{a} = -8$ $\quad -56x = -104 \pmod{19}$

$19 = 2 \cdot 7 + 5$

$7 = 1 \cdot 5 + 2$

$5 = 2 \cdot 2 + 1$

$2 = 2 \cdot 1 + 0$

$1 = 5 - 2 \cdot 2$

$1 = 5 - 2(7 - 1 \cdot 5) = 3 \cdot 5 - 2 \cdot 7$

$1 = 3 \cdot (19 - 2 \cdot 7) - 2 \cdot 7$

$1 = 3 \cdot 19 - 8 \cdot 7$

Bézout coef.: 3, (-8)

(7.) $ax + by = 1 \quad a = 5n+3 \quad b = 7n+4$

$(5n+3)x + (7n+4)y = 1$

$(7n+4) = (5n+3) + (2n-1)$

$5n+3 = 2(2n-1) + (n+1)$

$2n+1 = (n+1) + n$

$n+1 = n + 1$

$1 = (n+1) - n$

$1 = (n+1) - [(2n+1) - (n+1)]$

$1 = 2(n+1) - (2n+1)$

$1 = 2[(5n+3) - 2(2n+1)] - (2n+1)$

$1 = 2(5n+3) - 5(2n+1)$

$1 = 2(5n+3) - 5[(7n+4) - (5n+3)]$

$1 = 7(5n+3) - 5(7n+4)$

$(x, y) = (7, 5)$

(8)